

AfriVid Studio

Phase 2: Server-Side Video Processing

Full Technical & Business Documentation | Version 1.0 | June 2026 | Confidential

Executive Summary

Phase 2 transforms AfriVid Studio from a browser-based editor into a full server-side AI video processing platform. Users can upload videos of any length, submit processing jobs, and receive professionally edited videos via email — without their browser freezing or crashing.

Item	Details
Investment	\$22 one-time + \$12-20/month running costs
Timeline	4-6 weeks to build
Capacity	500 Pro users fluently from day 1
Revenue Potential	\$2,500/month at 500 Pro users
Break-even	5 Pro users at \$5/month

1. The Problem Phase 2 Solves

Current Limitation	Impact on Users
Max ~5 minute videos	Churches, organizations, YouTubers cannot use AfriVid
Browser freezes on long videos	Users frustrated and leave
Must keep browser tab open	Cannot close laptop during processing
Limited by device RAM	Low-end phones cannot process anything
No file storage	Videos lost after browser closes
No background processing	User stuck waiting and staring

2. What Phase 2 Fixes

Feature	Before Phase 2	After Phase 2
Video length	~5 min max	Up to 3 hours
Processing	Browser (slow, crashes)	Server (fast, reliable)
User experience	Wait staring at screen	Submit and leave
Device requirement	Powerful laptop only	Any phone works
Storage	None — lost on close	R2 cloud storage
Notification	None	Email when done
Concurrent users	1 at a time	100+ in queue

3. Technical Architecture

System Components:

Component	Purpose
Cloudflare Worker	API gateway — routes requests, validates users, creates jobs
Cloudflare R2	Video storage — input videos + processed output + thumbnails
Fly.io Server	Processing engine — FFmpeg + Python scripts (2GB RAM)
Replicate AI	Whisper (transcription), Demucs (noise removal), RMBG (bg removal)
Claude API	Highlight detection, transcript analysis, script generation
Firebase Firestore	Job queue, status tracking, progress updates
Resend Email	Notify users when processing starts and completes

4. Processing Pipeline

1. User selects video file (any length, any size)
2. Browser uploads to Cloudflare Worker /upload-video
3. Worker validates user authentication and plan
4. Video streamed and stored in R2 bucket
5. Job created in Firestore with unique job_id
6. User sees: 'You are #3 in queue. Estimated: 15 minutes'
7. Fly.io server polls Firestore every 30 seconds for new jobs
8. Server downloads video from R2 and begins processing
9. Whisper transcribes audio with timestamps
10. Claude analyzes transcript to find best moments
11. FFmpeg cuts silences, creates highlight reel, trims video
12. Demucs removes background noise from audio
13. Google TTS generates translated audio if requested
14. FFmpeg renders final output video
15. Output uploaded to R2 with public download URL
16. Email sent to user: 'Your AfriVid is ready! Download here'
17. User downloads professionally edited video

5. Processing Features & Costs

Feature	Technology	Cost per Video
Transcription	Whisper Large V3 via Replicate	\$0.06
Silence Removal	FFmpeg VAD filter	Free
Highlight Reel	Claude API + FFmpeg	\$0.02
Voice Translation	Whisper + Google TTS	\$0.04
Noise Removal	Demucs via Replicate	\$0.10

Background Removal	RMBG-2.0 via Replicate	\$0.003
Thumbnail Generation	Flux AI (Cloudflare)	Free
Total per 1-hour video		~\$0.22

6. Cost Breakdown

One-Time Setup:

Item	Cost
Replicate testing credits	\$10
Fly.io first month (2GB RAM)	\$7
Cloudflare Workers Paid first month	\$5
TOTAL ONE-TIME	\$22

Monthly Running Costs:

Service	Monthly Cost
Fly.io 2GB RAM server	\$7
Cloudflare Workers Paid	\$5
Per video processing (~\$0.22/video)	Variable
Resend emails	Free
R2 Storage (first 10GB)	Free
TOTAL FIXED	\$12/month

7. Revenue vs Cost Projection

Pro Users	Monthly Revenue	Total Cost	Monthly Profit
5	\$25	\$13.10	\$11.90
10	\$50	\$14.20	\$35.80
20	\$100	\$16.40	\$83.60
50	\$250	\$23.00	\$227.00
100	\$500	\$34.00	\$466.00
500	\$2,500	\$122.00	\$2,378.00

8. Server Capacity

Video Length	Processing Time	Daily Capacity	Monthly Capacity
5 minutes	~2 minutes	~200 videos	~6,000 videos
30 minutes	~10 minutes	~50 videos	~1,500 videos
1 hour	~20 minutes	~25 videos	~750 videos
Mixed average	~8 minutes	~80 videos	~2,400 videos

At 500 Pro users processing 5 videos/month = 2,500 videos/month. Daily capacity of ~80 videos = 2,400/month. Handles 500 users fluently. ■

9. Build Timeline

Week	Tasks	Deliverable
Week 1	Fly.io setup, R2 upload endpoint, job queue	Video uploads to server
Week 2	FFmpeg silence removal, trim, progress tracking	Basic processing works
Week 3	Whisper transcription, Claude highlights, translation	AI features work
Week 4	Frontend progress UI, email notifications, download	Full user experience
Week 5	Testing (5min, 30min, 1hr videos), bug fixes	Stable platform
Week 6	Beta launch, monitor, fix, public launch	Phase 2 live

10. Pages That Benefit from Phase 2

Page	Current State	After Phase 2
AI Video Editor	Max ~5 min videos	Up to 3 hours, all features
Video Creator	Max 2 min videos	Up to 60 min videos
Video Editor	Limited export quality	4K export, unlimited combine
Photo Editor	Basic BG removal	Professional RMBG-2.0 AI
Design Studio	One image at a time	Batch processing, better quality
My Studio	No real storage	Full R2 storage + streaming
Transcribe (NEW)	Does not exist	Upload audio → full transcript
Dubbing (NEW)	Does not exist	Full video dubbing in African languages
Podcast (NEW)	Does not exist	Audio → video with captions

11. Risk Assessment

Risk	Likelihood	Mitigation
Server crashes	Medium	Auto-restart + health checks every 30s
Large file uploads (>2GB)	Low	2GB upload limit enforced
Slow processing queue	Medium	Email notification hides wait time
Unexpected cloud charges	Low	Spending limits set on all services
Replicate API downtime	Low	Fallback to basic FFmpeg processing

12. New API Endpoints

Endpoint	Method	Purpose
/upload-video	POST	Upload video to R2, create processing job
/job-status/:id	GET	Check job progress (0-100%)

/my-jobs/:uid	GET	Get user job history
/job/:id	DELETE	Cancel a queued job
/download/:id	GET	Get signed download URL from R2

13. Future Expansions (Phase 3)

- AfriVid Transcribe — Standalone transcription tool for journalists and students
- AfriVid Dubbing — Full video dubbing with lip sync in African languages
- AfriVid Podcast — Convert audio podcasts to video with captions
- Batch Processing — Process multiple videos simultaneously
- AfriVid API — Developers can integrate AfriVid processing in their apps
- Mobile App — React Native app backed by Phase 2 server
- AfriBrain Integration — Replace Claude API with AfriBrain model

14. Success Metrics (Month 1 Targets)

Metric	Target
Videos processed	50+
Average processing time	< 20 minutes
Processing success rate	> 90%
Pro users gained	5+
Monthly revenue	\$25+
User satisfaction rating	> 4/5 stars

Regan Odhiambo Ayiecho — Founder & CEO, AfriVid Studio
Email: odhiamboregan6440@gmail.com | afrivid.studio | github.com/rayiecho
ALU Kigali | Mastercard Foundation Scholar | Age 21

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